GEHA DENTAL RAG · GENERATOR FINETUNING

Teaching the answer model to read our passages

LoRA instruction finetuning of Qwen2.5-1.5B-Instruct on direct and chain-of-thought examples

October 2026 · experiment log · results as of 21 Oct, 07:34

Two places to finetune a RAG system

RETRIEVER · DONE EARLIER

Which chunks the LLM sees

bge-base-en-v1.5 embedding model, contrastive loss on question → chunk pairs.

Recall@5: 64.6% → **68.7%** on the held-out brochure

GENERATOR · THIS DECK

How the LLM answers from them

Qwen2.5-1.5B-Instruct chat model, LoRA instruction finetuning on passage + question → answer.

Validation loss: 0.626 → **0.517** after one epoch

The slide that started this: instruction finetuning with direct and chain-of-thought data (FLAN, arXiv 2210.11416)

The pipeline, end to end

01

QA pairs

Synthetic RAGAS questions + reference answers from the GEHA dental PDFs

→

02

SFT builder

Gold passage + 2 distractors, shuffled. One direct and one chain-of-thought target each

→

03

LoRA training

Loss on answer tokens only. Keep the adapter with the lowest validation loss

→

04

Held-out eval

Unseen plan brochure. Base vs tuned, same prompts; only the adapter differs

Why distractors: retrieval is never perfect, so the model must learn to ignore irrelevant chunks (the RAFT recipe).

Why two formats: mixing direct and chain-of-thought targets in one run is the FLAN multi-task recipe.

`prepare_sft.py → train_lora.py → plot_loss.py → eval_generator.py`

One question becomes two training examples

PROMPT · masked, not trained on

[1] guide p.6 · Plan features (distractor)
[2] guide p.2 · Contact us (gold)
[3] guide p.4 · Both plans include (distractor)

Question: What is the phone number to contact a FedViser for GEHA dental benefits?

DIRECT TARGET · trained on

You can reach a FedViser … by calling 1-855-572-1636 Monday through Friday from 8 a.m. to 8 p.m. Eastern time.

CHAIN-OF-THOUGHT TARGET · trained on

Evidence:
- [2] “Reach a FedViser Monday-Friday from 8 a.m.-8 p.m.”
Answer: You can reach a FedViser … 1-855-572-1636 …

Real example ragas-0002 (abridged). Evidence is extractive: the gold sentences that overlap the reference answer most.

Data splits: the test set is a different document

	train	validation	test
Document	guide	guide	brochure
Sections	22	6	72
QA pairs	78	24	41
SFT examples	156	48	82
Distractor pool	34	8	167
Prompt tokens, mean	990	702	1,251
Answer tokens, mean	135	120	199

Leakage checks: 5 of 5 pass

- No shared chunk IDs
- Train/val sections disjoint
- Test document unseen
- No repeated questions
- Sources inside own split

Guide = 2026 GEHA dental benefits guide · Brochure = 2026 plan brochure · Report: results/data_splits.md

The model: a frozen 1.5B base with a small adapter

1.5B
frozen weights in Qwen2.5-1.5B-Instruct

18.5M
trainable LoRA weights, about 1.2% of the model

20
optimizer steps per epoch (156 examples ÷ batch 8)

74 MB
adapter file on disk, vs about 3 GB for the base

LoRA	Targets	Optimizer	Selection
rank 16, alpha 32, dropout 0.05	q, k, v, o, gate, up, down	AdamW 2e-4, warmup + cosine	lowest validation loss

Adapter size measured from adapter_model.safetensors (fp32); base size is the bf16 checkpoint

On the Mac: 4 h 41 min, one finished epoch

02:53

Start on an M1 Pro, 32 GB, fp32 on the GPU (MPS)

03:57

Epoch 1 done in 56 min. Validation loss 0.626 → 0.517; adapter saved

06:39

Battery at 1%: the Mac hibernates for 12 min, epoch 2 still unfinished

07:34

Stopped. Full run moves to a Colab GPU

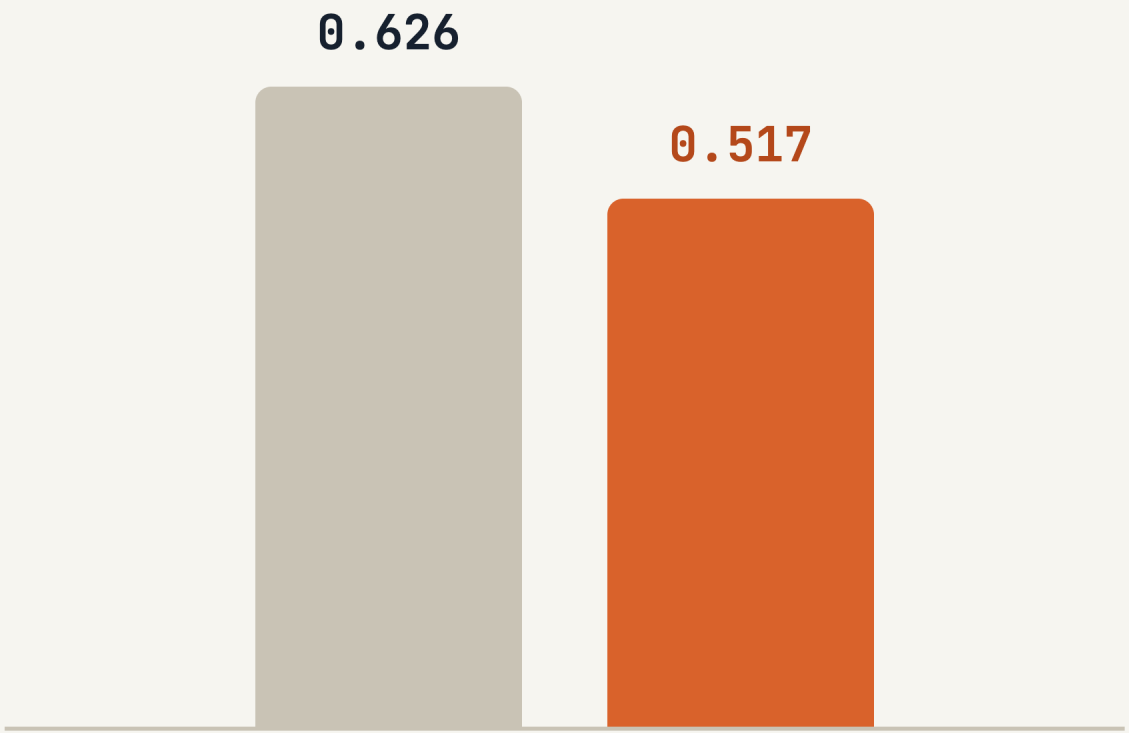
No bf16 on M1. fp32 doubles memory and compute compared with a bf16 GPU.

Ran on battery. macOS throttled the job, then hibernated it.

Memory pressure. 15.8 of 17 GB swap in use slowed every step.

Times are local (PDT), 21 Oct 2026, from the training log and macOS power log

Performance so far: validation loss down 17% in one epoch



Before training (grey): 0.626 average loss on the reference answers of 48 validation examples.

After epoch 1 (orange): 0.517. The tuned model assigns noticeably higher probability to the reference answers.

Not yet measured: answer quality on the unseen brochure (answer F1, citations, grounded quotes). That comes from the Colab run.

Validation = 24 questions × 2 formats from guide sections never trained on · bars start at zero

Is it overfit? Not after epoch 1, but that’s not proven yet

WHAT WE SEE

```
train 0.517
val   0.517
```

No gap between training and held-out loss, and validation loss is still falling. The usual sign of overfitting, train down and validation up, has not appeared.

WHY IT’S NOT SETTLED

Only 20 steps so far; overfitting usually shows in later epochs.

Validation is the same guide as training, so it can’t show transfer to new documents.

78 synthetic QAs can teach RAGAS answer style rather than reading skill.

WHAT WOULD SHOW IT

Validation loss rising in epochs 2–3.

Brochure answer F1 at or below the base model.

More citations to distractors, or quotes not found in the passages.

Train loss is the epoch average, so end-of-epoch train loss is likely a little lower than 0.517

Next: the full run on a Colab GPU

1. **Bundle:** make_colab_zip.sh packs code + data (0.9 MB)
2. **Open** colab_train.ipynb on an L4 or A100 GPU (bf16)
3. **Run:** tests, split report, 3 epochs with loss logged every step, eval
4. **Download** results + adapter; this deck gets the real numbers

To fill in from that run

Measure	Value
GPU and time per epoch	[__]
Best validation loss (epoch)	[__]
Answer F1, base → tuned	[__]
Citation precision (CoT)	[__]
Grounded quotes (CoT)	[__]

`finetuning_LLM_LORA/ · README.md has every step and flag`